Beyond the Smoke: Investigating the Impacts of Cannabis Legalization on Societal Drug Patterns in Canada

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This study investigates the effects of cannabis legalization in Canada on the consumption of cannabis, alcohol, and hard drugs. We analyzed cross-sectional micro-data from Statistics Canada surveys conducted before and after legalization in 2018. Our findings reveal a significant increase in cannabis consumption post-legalization, but no significant rise in hard drugs, challenging the assumption that legalization leads to a broader permissive environment for drug use. Interestingly, there's a notable shift in alcohol consumption patterns post-legalization. The study offers insights into the complex interplay of drug consumption behaviors in response to policy changes, providing valuable information for policymakers and public health professionals in understanding and addressing the broader implications of cannabis legalization.

Legalization | Cannabis | Drugs | Canada

The legalization of non-medical and recreational cannabis in Canada in October 2018 marked a significant paradigm shift in drug policy, being only the second country to do so after Uruguay. The Cannabis Act grants the federal government responsibility for various aspects, such as production, licensing, and health warnings, while provinces handle retail and have the liberty to regulate age limits, possession amounts, and cultivation requirements. The purpose of the Cannabis Act encompasses protecting youth health, preventing illicit activities, reducing the criminal justice burden, ensuring a regulated and quality controlled cannabis supply(1).

This research paper aims to investigate whether the legalization in Canada encourages the consumption of cannabis and generates a spillover effect on the consumption pattern of cannabis and, more interestingly, other substances, such as alcohol and hard drugs, within the Canadian population. Particularly, we would like to test if cannabis legalization results in a significant increase in consumption of other substances. Our research question sparked from the widely diffused belief, often raised in the public debate, that legalization creates a perceived more permissive environment that might lead to a rise in the usage of other substances. To do so, in this paper, a binomial logistic regression model is employed, using repetitive cross-sectional micro-data from two surveys conducted by Statistics Canada in 2017 and 2019. The findings of this research might provide policymakers and the public opinion with a deeper understanding of the effects of cannabis legalization.

Our research question traces back its roots in existing literature that have explored the effects of legalization of cannabis on cannabis itself, alcohol, and hard drug consumption. Some authors focused their research mainly on legalization effects on cannabis. Although the overall outcome is predictable (2), that is, the usage of this substance increases, several interesting results have been found. For instance, the literature finds that men have always been more likely to use cannabis than women, but after legalization the gender gap of cannabis usage narrows (3). There is also evidence that legalization affects more adult than youth in relation to cannabis consumption (4), (5). These findings are consistent with the report published in October 2023 by Statistics Canada that summarizes how Canada's use of cannabis developed since its legalization (6). Within all possible factors that have impacted cannabis consumption after legalization, sociological ones, such as social acceptance, are the most relevant for the purpose of this paper. For instance, it was found that legalization is likely to affect positively the usage of the legalized substance by reducing the stigma of illegality (7). This is particularly relevant in our analysis since one typical assumption in public debate is that the perceived illegality reduces for every drug, generating a more liberal environment.

To regard with alcohol instead, the assessment of the impact of legalization becomes more intricate. This is mainly due to the fact that the joint behaviour of cannabis and alcohol is not well conclusive. In some papers a substitution effect is found, while others argue a positive correlation. This issue is reflected in the correspondent research that highlights how the policy characteristics, such as policy longevity, might impact the drugs consumption pattern. (8), (9).

Finally and more interestingly for our purposes, some authors focused their research on the effect of legalization on hard drugs. Past research shows a positive association between cannabis use and hard drugs, while more recent literature seems to suggest that this anticipated result might fail to be supported by data. (5), (2). For instance, it was found that cocaine and heroin consumption does not increase significantly after cannabis legalization (10).

Result

In light of Cannabis Act on October 17, 2018, we have undertaken a comparative analysis of drug consumption data in 2017 (pre-legalization) and 2019 (the year following legalization). This approach allows for the creation of a binary variable to assess the impact of the *Treatment*. Using a binomial logistic regression to examine the impact of different factors on cannabis use, it was found that the legalization of cannabis has a statistically significant effect. As the table 1 shows, the probability of cannabis consumption is doubled post-legalization compared to pre-legalization, as the coefficient associated with the *Treatment* equal to 2.027 demonstrates. This initial finding

Table 1. Treatment and its Interaction with Sex Outcomes.

	Coefficients of independent variables:		
	Treatment	Sex Pre-Treat	Sex Post-Treat
CAN	2.027***	0.675***	1.063$^{\cdot}$
	(0.399)	(0.024)	(0.037)
ALC	1.483$^{\cdot}$	0.811***	1.079$^{\cdot}$
	(0.245)	(0.038)	(0.050)
COC	0.948	0.513***	1.077
	(0.247)	(0.027)	(0.056)
HER	1.519	0.416***	1.206
	(1.242)	(0.092)	(0.268)
MET	1.045	0.540***	1.079
	(0.395)	(0.046)	(0.093)
SAL	0.390	0.331***	1.333
	(0.563)	(0.036)	(0.147)
XTC	0.543	0.748***	1.064
	(0.344)	(0.045)	(0.065)
HAL	1.076	0.516***	0.933
	(0.238)	(0.022)	(0.040)
GLU	0.892	0.420***	1.276
	(0.550)	(0.066)	(0.200)

Note: $^{\cdot}p < 0.1$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

This table illustrates coefficients and SE's from logistic regressions. In all models, except for Cannabis and Alcohol, we can reject the Null HP. that *Treatment* is significant with p < 0.10. We cannot reject the Null HP. that Sex Pre_Treat is significant with p < 0.10.

aligns with established theories, consistently demonstrating an increase in consumption following legalization.

Furthermore, before legalization, the probability of consuming cannabis is lower in women compared to men as it is shown by a lower than 1 odds ratio, while after treatment, the likelihood of cannabis consumption is higher among women than men, as exhibited by an associated odds ratio higher than 1. This outcome suggests that, overall, cannabis is more likely to be consumed by men, but legalization contributes to a reduction in the gender gap in consumption. Our result is consistent with (3), since we observed the same pattern in terms of narrowing gender gap. Additionally, it is noteworthy that the probability of cannabis consumption decreases incrementally for individuals with a self-declared higher mental health status. For instance, an individual with an excellent mental health status has an 85% lower probability of cannabis consumption compared to someone with a very low mental health status, as displayed in Fig. 1.

Considering the age of consumers, from our analysis is emerges that, pre-legalization, the probability of cannabis consumption increases with age until the 23-24 age bracket, gradually decreasing as individuals become older. This trend culminates in an 82% lower probability of cannabis consumption among those over 65 compared to the 15-17 age group, as Fig.1 shows.

Subsequently, we explored the legalization's impact on the consumption of other hard drugs, testing the hypothesis that a more permissive drug environment may lead to a general increase in consumption. Analyzing various binomial logistic regressions for different types of substances (cocaine, hallucinogens, methamphetamine, ecstasy, heroin, stimulants, salvia, solvents, and alcohol) using demographic variables and the *Treatment* variable as covariates, we found no significant effect of legalization on any heavy drug, as Fig. 2 confirms. The only substance exhibiting a significant effect, other than cannabis, is alcohol, with an increase in consumption probabilities post-cannabis legalization of 48,3%, a factor worth considering in designing legalization policies.

Lastly, evaluating interaction variables, a common pattern emerges across all tested heavy drugs. Post-treatment, despite non significance, there is a pattern for the probability of drug consumption to increase among women compared to men, corroborating the previous result on cannabis. Table 1 includes the coefficients of all drugs.

Discussion

As expected, our model shows that cannabis consumption increases significantly as a result of cannabis legalization, a largely anticipated outcome that can be explained by lower prices, ease of purchasing and social acceptance. More interestingly, the model reveals that there is not sufficient statistical evidence to state that consumption of heavy drugs, such as cocaine, methamphetamine or heroin, witnesses any significant variation as a result of the legalization. Hence, from our analysis, it seems that legalization has not worked as a "free-for-all", that is, a social environment that lead to an uncontrolled substance use. These findings challenge the common argument in public debates that legalization fosters a permissive environment that encourage drug use in general.

However, our model has identified some critical issues associated with the implementation of recreational cannabis. For instance, one possible drawback of legalization, highlighted by our research, relates to the behaviour toward alcohol consumption. Indeed, individuals seem to be more prone to consume alcohol after cannabis legalization, as it is proved by the positive significant variation (at 10%) shown by the model. One possible solution, in order to avoid this pattern, might be to increase taxes on alcohol beverages, as the government implements cannabis legalization laws. A contraction in alcohol consumption, however, might impact and generate unknown effects on the consumption of cannabis and/or heavier drugs. Studies about whether these substance are substitutes are not conclusive and our research does not provide any significant information to corroborate or reject one possibility or the

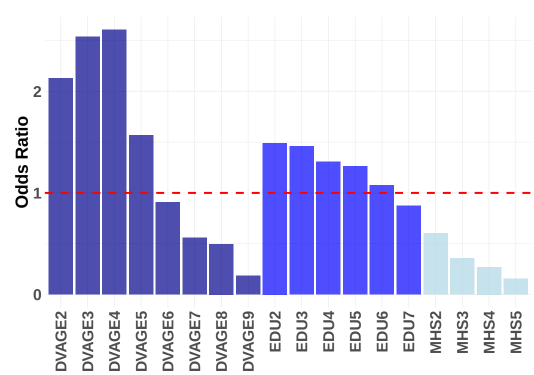

Fig. 1. Odds ratio values for CAN model.

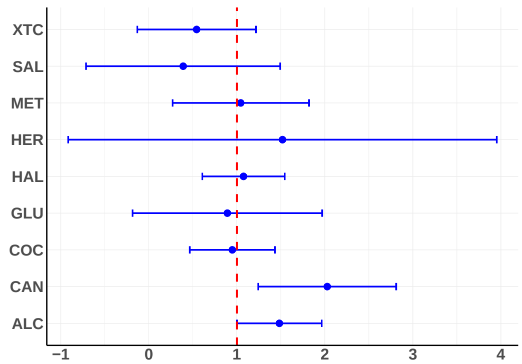

Fig. 2. Treatment odds ratio of each substance with Estimates and Standard Error.

other. As a result, deeper analysis should be conducted in order to evaluate this kind of proposal, but still considering that legalization might lead to higher alcohol consumption.

Limitations

In this study, there are factors constraining our ability to generalize the results, some intrinsically related to the survey and others to our model. Firstly, as we are going to discuss in greater details later, both 2017 and 2019 surveys were telephone based. The choice is prone to error due to missing partial responses, particularly considering the sensitive questions that are asked. To address this issue, we excluded all the variables with a high skip rate. Nevertheless, the exclusion of potentially relevant variables represents itself a constraint, since it narrows the possible combinations we could have explored.
Moreover, we decided to exclude more recent surveys due to availability and reliability issues. Indeed, StanCan have not made available micro-data about 2021 and 2023 inquiries, but only aggregate variables and general patterns that are incompatible with our framework. Additionally, the result have been almost certainly influenced by the Covid-19 pandemic. A potential research approach that could have been applied if these missing surveys were available is to assess the so-called gateway theory, which necessitates more cross-sectional observations and involves long-term evaluation and analysis.
Future research could also include a Difference-in-Differences (DID) analysis, comparing provinces in Canada where cannabis was legalized (treated group) with neighboring regions where it remained illegal (control group). To ensure a robust analysis, it's important to select control regions with similar socio-demographic characteristics and geographical proximity, such as United States northern region. We excluded this framework, because of the fragmentation of cannabis policies across the US.

Materials and Methods

Data. We use cross-sectional micro-data from two surveys (2017, 2019) conducted by Health Canada, a federal institution, in partnership with StatCan. These surveys are the latest available additions to the ongoing series of biennial general population surveys that examine alcohol and drug use among Canadians aged 15 years and older. These two surveys are particularly interesting since data were collected just before and right after cannabis legalization. Both of them are telephone based, as data were collected using computer-assisted telephone interviews. The target population was all persons 15 years of age and over living in Canada with the exception of remote territories. Both geographical aspects as well as the composition of households were employed as criteria in a two-phase stratification of the survey. During the first phase, households are chosen through a random selection process from a random sample of households across the province-strata. In the subsequent phase, one or two individuals, or none, are then selected based on the composition of the household with the intent of collecting more target age group data. After a data set cleaning procedure, it encompasses 26642 observations and 21 variables. All of them are categorical variables with 2 or more levels. For instance, the main variable is "CAN" that takes value '1' if the respondent never used Cannabis in the 12 months preceding the survey, instead, value '2' if the respondent consumed it.

Model. To investigate the impact of the 2018 legalization of cannabis in Canada on the consumption of various illegal substances, we employed a binomial multiple logistic regression model, due to the binary nature of the dependent variable and the categorical nature of the regressors. This analytical approach allowed for the assessment of the significance of odds ratios, elucidating the likelihood of increased or decreased probabilities of response variable, given the isolated impact of the regressor. The model incorporated interaction terms, specifically the product of the *Treatment* variable, which filters data based on the pre-legalization year (2017) and the post-legalization year (2019), with various demographic variables. These interaction terms were introduced to assess whether the relationship between the response variable and the independent variables changes given the legalization. For illustrative purposes, we present the structure of an exemplary model focused on cannabis consumption:

$$\mathbf{Can} = \beta_0 + \beta_1 \times \mathbf{treatment} + \beta_2 \times \mathbf{sex} + \beta_3 \times \mathbf{treat\ \ sex} + \beta_4 \times \mathbf{edu} + \beta_5 \times \mathbf{treat\_edu} + \beta_6 \times \mathbf{dvage} + \beta_7 \times \mathbf{treat\_dvage} + \beta_8 \times \mathbf{dvurban} + \beta_9 \times \mathbf{treat\_dvurban} + \beta_{10} \times \mathbf{mhs} + \beta_{11} \times \mathbf{treat\_mhs}.$$

This modeling framework was replicated for each hard drug, analyzed as response variable, maintaining consistency across the investigation.

1. Branch LS (2023) Consolidated federal laws of canada, cannabis act (https://laws-lois.justice.gc.ca/eng/acts/c-24.5/page-1.html#h-76878).
2. Anderson DM, Rees DI (2014) The legalization of recreational marijuana: How likely is the worst-case scenario? *Journal of Policy Analysis and Management* 33(1):221–232.
3. Matheson J, Le Foll B (2023) Impacts of recreational cannabis legalization on use and harms: A narrative review of sex/gender differences. *Frontiers in Psychiatry* 14:1127660.
4. Hall W, Lynskey M (2020) Assessing the public health impacts of legalizing recreational cannabis use: The us experience. *World Psychiatry* 19(2):179–186.
5. Smart R, Liccardo Pacula R (2019) Early evidence of the impact of cannabis legalization on cannabis use, cannabis use disorder, and the use of other substances: Findings from state policy evaluations. *The American Journal of Drug and Alcohol Abuse* 45(6):644–663.
6. Statistics Canada (2023) Five years since legalization, what have we learned about cannabis in canada? (https://www150.statcan.gc.ca/n1/daily-quotidien/231016/dq231016c-eng.pdf).
7. Jacobi L, Sovinsky M (2016) Marijuana on main street? estimating demand in markets with limited access. *American Economic Review* 106(8):2009–2045.
8. Gunn RL, Aston ER, Metrik J (2022) Patterns of cannabis and alcohol co-use: Substitution versus complementary effects. *Alcohol Research : Current Reviews* 42(1):04.
9. Guttmannova K, et al. (2016) Impacts of changing marijuana policies on alcohol use in the united states. *Alcoholism: Clinical and Experimental Research* 40(1):33–46.
10. Chu YWL (2015) Do medical marijuana laws increase hard drug use? *Journal of Law and Economics* 58(2):1–60.